

RED BLUFF, CA, USA

WMO: 725910

Lat: 40.152N

Lon: 122.254W

Elev: 353

StdP: 14.51

Time Zone: -8.00 (NAP)

Period: 95-19

WBAN: 24216

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	30.6	33.3	12.1	10.3	56.4	16.5	12.8	56.6	34.5	52.7	30.0	50.7	4.9	320	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.7	105.1	68.3	101.7	67.3	98.9	66.7	72.5	95.0	70.8	93.2	69.3	91.6	8.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.2	94.3	81.6	63.2	88.0	80.4	61.2	81.8	78.9	36.2	95.6	34.8	93.0	33.5	91.9	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.0	22.1	18.7	DB	25.6	111.2	2.9	2.4	23.5	113.0	21.7	114.4	20.1	115.8	18.0	117.5	
			WB	23.8	75.8	3.3	2.0	21.4	77.2	19.5	78.4	17.6	79.5	15.2	81.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.5	47.6	50.7	55.0	59.7	68.0	77.0	82.3	80.1	75.0	64.9	53.7	47.2	(6)
	DBStd	13.83	5.28	5.60	6.23	7.15	7.59	7.41	5.28	4.89	6.41	6.39	6.19	5.34	(7)
	HDD50	336	109	53	21	6	0	0	0	0	0	0	31	116	(8)
	HDD65	2483	539	400	314	191	55	6	0	0	5	81	341	551	(9)
	CDD50	5268	35	73	177	296	559	809	1000	934	751	461	143	30	(10)
	CDD65	1941	0	0	5	32	149	365	535	469	306	77	3	0	(11)
	CDH74	26682	5	14	125	530	2082	4996	7642	6292	3882	1049	65	0	(12)
	CDH80	15219	0	0	19	174	976	2883	4787	3813	2169	389	9	0	(13)
(14) Wind	WSAvg	7.7	7.1	8.8	8.7	8.5	8.3	8.4	6.8	6.4	6.9	7.4	7.0	7.8	(14)
(15) Precipitation	PrecAvg	22.6	4.1	3.6	3.0	1.4	0.8	0.4	0.1	0.2	0.5	1.3	3.3	3.7	(15)
	PrecMax	49.0	10.2	8.6	9.2	4.9	3.3	1.6	0.7	1.6	5.0	5.2	8.4	10.3	(16)
	PrecMin	7.2	0.2	0.0	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.0	(17)
	PrecStd	7.6	2.5	2.8	1.9	1.3	0.9	0.5	0.2	0.4	0.9	1.4	2.4	2.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.5	76.4	82.6	90.1	99.4	108.4	109.9	108.0	104.4	94.4	80.9	69.1	(19)
		MCWB	52.0	56.0	58.5	62.3	66.4	68.5	69.6	69.9	65.7	62.7	57.7	50.7	(20)
	2%	DB	66.8	70.7	77.4	85.1	94.3	102.8	105.3	103.1	99.4	88.4	74.9	63.3	(21)
		MCWB	49.8	53.2	57.1	60.2	64.5	67.1	69.0	67.4	65.0	60.4	56.1	49.3	(22)
	5%	DB	61.4	65.6	73.0	80.6	90.1	98.5	102.0	99.7	96.1	84.3	70.2	59.4	(23)
		MCWB	49.4	51.0	55.6	59.0	63.1	66.1	68.0	67.2	64.0	59.1	54.4	49.4	(24)
	10%	DB	56.8	61.4	68.3	75.3	85.4	94.3	98.9	96.7	92.4	80.0	65.2	56.5	(25)
		MCWB	48.5	50.3	53.7	56.8	61.3	64.8	67.7	66.0	63.3	57.6	52.6	49.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.5	58.5	61.7	65.5	70.0	74.4	74.5	75.5	71.3	65.3	61.3	57.4	(27)
		MCDB	61.2	67.1	74.6	82.2	90.7	95.9	97.1	98.8	91.7	85.7	73.9	60.0	(28)
	2%	WB	53.7	56.3	59.3	62.5	67.3	71.0	72.7	71.6	68.2	63.1	58.8	55.0	(29)
		MCDB	58.3	64.6	72.2	79.6	88.3	92.5	95.9	93.5	89.6	82.7	68.5	57.3	(30)
	5%	WB	52.2	54.2	57.4	60.4	65.1	68.8	71.1	69.7	66.4	61.4	56.8	53.1	(31)
		MCDB	57.0	61.6	68.6	76.2	85.4	90.5	94.2	91.4	88.9	78.4	65.0	55.7	(32)
	10%	WB	50.7	52.3	55.4	58.4	63.0	67.1	69.7	68.1	64.8	59.6	55.0	51.1	(33)
		MCDB	55.8	59.1	65.6	72.0	81.7	89.4	92.9	90.2	87.8	75.1	62.3	54.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.1	18.4	21.0	23.2	25.7	27.6	30.7	31.1	30.2	26.0	20.1	16.5	(35)
		MCDBR	27.8	27.7	29.6	31.8	33.2	33.1	34.6	35.0	35.3	33.0	28.8	23.9	(36)
	5% WB	MCWBR	14.8	14.1	14.3	13.6	12.3	11.2	11.1	11.9	12.5	13.5	13.8	13.6	(37)
		MCWBR	17.9	20.4	25.4	29.1	30.7	30.3	31.1	31.1	31.0	28.0	22.3	12.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.314	0.316	0.336	0.351	0.341	0.349	0.354	0.337	0.339	0.329	0.310	0.314	(40)
		taud	2.528	2.530	2.478	2.428	2.497	2.478	2.452	2.494	2.480	2.522	2.553	2.527	(41)
	Ebn at Noon		271	288	292	293	297	293	290	294	287	279	271	262	(42)
		Edh at Noon	23	27	31	35	34	34	35	33	31	27	23	22	(43)
(44) All-Sky Solar Radiation	RadAvg		658	949	1346	1833	2258	2572	2608	2320	1855	1262	774	557	(44)
	RadStd		121	135	169	184	161	140	82	80	75	108	81	111	(45)

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+1.06	N/A	-2.29	N/A	N/A	-74	-257	+364	+202
(47)	Regional (23 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	+209	+49

Nomenclature: See separate page